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Enhancing accuracy through ensemble based machine learning for intrusion detection and privacy preservation over the network of smart cities

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Abstract
The world is at peak of fifth-generation communication technology and adopting ideas like cloudification or virtualization, but, the most important element is still "security", since more and more data is connected to the internet. Threat attacks are increasing in recent years, but classic network intrusion detection system has significant limitations that make it challenging to identify new attacks quickly. In this study, various supervised machine learning algorithms for anomaly-based detection methods are compared. The dataset utilized for anomaly-based detection techniques is KDDCup99 dataset, on which the different algorithms have been applied. The goal is to gain knowledge about data integrity and improve the predictive power of data. Given its ability to safeguard the integrity of data storage and maintain transparency in processes, this technology has promise for application in the field of intrusion detection. This study presents a technique for measuring various data parameters in network like accuracy, error rate, confusion matrix, etc. By accomplishing this, the amount of malicious data floating around in network can be reduced, making it a safe environment for data sharing. The accuracy of the proposed technique was found to be 99.82% accurate using the KDDCup99 dataset.

Article Highlights

- This study demonstrates how machine learning can be used to overcome the drawbacks of conventional intrusion detection systems.
- Enhanced detection can drastically minimize malicious network activity, resulting in safer data sharing.
- Network intrusion detection accuracy is increased by 99.88% using ensemble machine learning techniques.

Keywords Intrusion detection system · eXtreme gradient boosting · KDDCup99 · Gaussian Naive Bayes · Decision tree · Max voting technique

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1 Introduction

Network intrusion detection is struggling more and more to guard against modern threat attacks. The standard feature-based intrusion detection system has been in use for a very long time. Machine Learning (ML) techniques improve the ability of intrusion detection systems to detect novel and sophisticated attacks that may not be covered by preconfigured signatures in a database. ML techniques can also address this limitation by enabling Intrusion Detection System (IDS) to learn from historical data and detect previously unseen attacks. ML techniques can identify patterns and anomalies in network traffic data, enabling the system to detect potential threats and raise alerts. One common approach is to use supervised learning techniques, where the algorithm is trained on labeled dataset of normal as well as malicious traffic to learn the characteristics of different attacks. Once the algorithm is trained, it can classify new network traffic as either malicious or normal. Unsupervised learning techniques, such as clustering and anomaly detection, can recognize unusual network behavior that may signify an attack. Overall, applying machine learning methods to intrusion detection systems can improve the ability of these systems to detect and prevent novel and sophisticated attacks, and thus enhance the security of computer systems and networks. Many improvements can be made in the field of intrusion detection by using ML techniques like decision trees, random forests, neural networks, and others in recent years. Network data traffic has risen as a result of technological development and information digitalization [1]. This increases the area of network attacks and puts data as well as computer systems at a risk. The use of internet appears to directly correlate with an increase in network security vulnerabilities. A network can be protected against intruders by using intrusion detection systems to identify illegal access and attacks [2].

A commonly used word in this regard is 'Intrusion', which means wrongfully gaining access to an individual's or organization's network to steal their data. There have been many attempts to prevent these intrusions, like firewalls, access control schemes, or encryption methods but none of them has proven to be effective in protecting the network and the systems from malware and attacks. Intrusion detection systems turns out to be an effective method that aims to detect intrusions and are a crucial component of any security infrastructure [3].

An intrusion detection system, abbreviated as IDS, is a predictive model that classifies network data traffic as normal or intrusion. IDS are of two types, namely misuse-based and anomaly-based. The misuse-based IDS locates an attack on the basis of attack signatures, hence it cannot detect any novel attacks. On the other hand, anomaly-based IDS detects an attack on the basis of recorded normal behavior that compares previous normal real-time traffic with actual real-time traffic. Intrusion detection systems are designed to detect and prevent unauthorized access to computer systems or networks. Traditional IDS systems rely on preconfigured signatures that match known attack patterns to identify and prevent attacks. However, these systems have limitations in detecting novel and sophisticated attacks, which can have serious consequences if they go undetected. This is a widely used type of IDS as it can detect novel attacks but it does register a large value of false positive alarm. There has been a lot of research to develop various machine learning models to cope with this problem in both types of IDS. A lot of work is involved in designing an IDS including data collection, pre-processing, developing models, and then classifying network traffic as intrusion and normal [4, 5]. The intrusion detection system assumes a pivotal role in providing comprehensive security protection by serving as the primary protective barrier. The findings obtained from intrusion detection systems have practical applications in various areas. They can be utilized to make self-adaptive judgments and provide real-time responses [6]. Additionally, IDS results can assist safety operators in promptly detecting aberrant states [7, 8].

The incursion of anomalies in Blockchain poses a substantial danger to the security and privacy of information. Consequently, it is imperative to build safe intrusion detection systems. Security detection algorithms for Blockchain data are extensively utilized in diverse domains of artificial intelligence [9]. However, it is important to note that the global financial system is vulnerable to security risks that have the potential to cause significant financial losses. Existing security methods, like identity authentication [10, 11], resource protection [12–15], and machine learning [16], have demonstrated their effectiveness in mitigating the security challenges associated with Blockchain. The existing intrusion detection techniques for Blockchain mostly focus on features of data characteristics, models of data characteristics, and collaborative detection.

Blockchain technology is a nascent solution that facilitates decentralized transactions and data management, eliminating the requirement for a trusted intermediary. The technology in question is a decentralized and transparent system that facilitates the documentation of transactions between several entities in a manner that can be authenticated. Until now, researchers have examined blockchain in many sectors such as healthcare and supply chain management. However, limited research has been conducted to explore the possible use of blockchain in the

realm of intrusion detection. In recent years, there has been a significant amount of interest in blockchain technology from both the academic and industrial sectors. This interest stems from the innovative nature of blockchain, which enables parties with mutual mistrust to exchange financial data without relying on a trusted third party. The aforementioned property is highly desirable for collaborative detection, as it presents an opportunity to address the challenges associated with data sharing and trust management [17, 18].

The examination of various intrusion detection solutions based on Blockchain [19] highlights the subsequent difficulties.

- The superfluous transaction information pertaining to intrusion detection ought to be diminished in order to mitigate the expenses associated with decentralization. The Blockchain is chosen as the preferred method for achieving optimal recognition of data clusters with high levels of security and decentralization. The issue lies in achieving the smallest number of clusters and consensus costs in intrusion detection, as the Blockchain necessitates a safe and energy-efficient verification of data transactions, which incurs resource expenses.
- The storage capacity of the intrusion detection algorithm is restricted, hence hindering its ability to reliably identify clusters within a given dataset under any circumstances. Hence, the algorithm in question is unable to accurately recognize clusters within a dataset that exhibit low-frequency fusion characteristics. Consequently, this limitation has a detrimental impact on the security of cross-chain technologies, specifically in the context of intrusion detection.
- Traditional approaches for detecting anomalies in networks have been shown to have limited accuracy and speed. Therefore, it is imperative to implement a high-speed and precise anomaly detection system in order to enhance the real-time management of a Blockchain network.

The main motivation behind the work is increased use of the internet that has increased the network data traffic which has paved the way for attackers to intrude into the network and steal sensitive information of any individual or organization. An efficient intrusion detection system is hence needed to build a strong model to classify data entering the network traffic accurately and prevent unauthorized access.

The current paper proposes an ensemble machine learning model that can integrate the benefits of each technique for various types of data detection and utilize ensemble learning to produce the best outcomes. By combining the predictions of numerous base estimators, ensemble machine learning has the advantage of increasing generalizability and robustness compared to using only one model. This study used KDDCup99 data set which is firstly cleaned up and made non-redundant through preprocessing before being tested against Decision Tree, Gaussian Naive Bayes, XGBoost and ensemble ML model.

The novelty of this research is in the proposal of an ensemble-based machine learning model aimed at improving the accuracy of intrusion detection systems. The use of ensemble machine learning model enhances the effectiveness of intrusion detection. The authors created an effective IDS and obtain good accuracy from the model. They outperform a lot of earlier study findings and offer promising practical possibilities. The significant contributions of this paper are summarized as follows:

- Applying pre-processing on the KDDCup99 dataset.
- Analyze the dataset through exploratory data analysis.
- Train the following classifiers: XGBoost, Gaussian Naive Bayes and Decision Tree.
- Proposed an ensemble-based machine learning model to enhance accuracy.

The rest of this paper is structured as follows: The background work on IDS is mentioned in Sect. 2. In Sect. 3, steps involved in various stages of development of the proposed model are presented. The experiment's findings and an ensemble machine learning technique is proposed in Sect. 4. The conclusion of the paper and future scope is discussed in last Sect. 5.

2 Background work

The current state of research on intrusion detection solutions for Blockchain-based systems is still in its early stages. Considering the fact that existing intrusion detection solutions for Blockchain networks impose significant costs for detecting intrusions in large networks, the process of monitoring the overall performance of the entire network results in

substantial communication costs and typically lacks promptness. Previous research has mostly concentrated on the topics of data consensus, completeness, privacy protection, and scalability in the context of Blockchain systems. Consequently, various data consensus algorithms have been presented. Nevertheless, the current anomaly detection algorithms result in significant computational expenses. The studies use various algorithms and have recorded commendable progress in terms of improving the performance of IDS.

The authors in paper [20] evaluated the efficiency of different ML techniques like Random Forest (RF), Decision Table (DT), Multilayer perceptron (MLP), Naive Bayes (NB), etc. against the KDD intrusion dataset and results demonstrated that the RF classifier achieved the maximum average accuracy rate while DT classifier obtained the minimum false negative. The study in [21] also evaluated the performance of various machine learning algorithms on the four categories of attack available in the KDD dataset and concluded that these algorithms did not offer many promising results in detecting U2R and R2L attacks. A similar study was done by [22] where the researchers tested SVM, Naive Bayes, Decision Table, and J-48 algorithm against the same benchmark KDD dataset. Their results concluded that each algorithm has different results for each class but the J-48 decision tree gave the highest accuracy of them all and a low misclassification rate. [23] researched in 2011 where the authors developed a lightweight IDS using a wrapper-based feature selection algorithm employing a neural ensemble decision tree procedure to maximize the sensitivity of the IDS. The study introduced an approach involving six classifiers namely Naive Baye's Tree, Random Forest, Representative Tree, Decision Stump, Random Tree and C4.5 model. The results justified that the proposed approach was a promising strategy to be applied to intrusion detection systems. Another approach was proposed by authors in [24] which reflected the efficiency of XGBoost in predicting a dataset. The model achieved an accuracy of 98.70% with a very low error rate. The findings of the model could then be elaborated to design an IDS. An ensemble classifier was proposed in [25] where the researchers combined 2 base classifiers to create an IDS and its future scope involves using another set of base classifiers to create an ensemble model with increased efficiency.

The authors have examined the NSLKDD dataset, which is excellent for evaluating various intrusion detection algorithms. Utilizing all 41 features in the dataset, to assess invasive patterns which may be time-consuming, but it will also prevent the system's performance from degrading. Therefore, this analysis demonstrates that Random Forest is accelerating the training and testing techniques for intrusion detection, that is crucial for the network application with highest speed and delivering the highest testing accuracy in the case of a limited feature set [26]. ANN is used in this study to assess the performance of the NSL-KDD dataset. The outcome for binary class classification and five class classifications (type of attack) are obtained. This paper describes several studies that have used the NSL-KDD dataset for intrusion detection and classification of different attack types. The studies have employed different machine learning techniques and performance metrics to evaluate the effectiveness of their models [27]. Another study suggests an adaptive ensemble learning model that uses the ensemble learning technique to improve the detection accuracy of intrusion detection systems. The study reports an accuracy of 85.2% for the proposed adaptive voting algorithm, which is higher than the accuracy of the MultiTree algorithm (84.2%). This model is shown to be effective in increasing detection accuracy compared to other research articles [28]. This paper analyzes the NSL-KDD dataset in terms of four classes: Content Basic, Host and Traffic. The analysis is conducted using 2 well-known performance metrics for IDS, namely, False Alarm Rate (FAR) and Detection Rate (DR). The contribution of qualities of each class on DR and FAR is displayed as a consequence of this empirical analysis of the dataset. The results of this study can improve the applicability of the dataset to achieve the highest DR with the lowest FAR [29]. Overall, these studies demonstrate the potential of ML methods to enhance the accuracy of IDS and provide insights into the evaluation metrics and dataset properties that can affect the performance of these systems. This study describes several studies that have used the NSL-KDD dataset for intrusion detection. The studies have employed different machine learning techniques and algorithms to improve the accuracy of the intrusion detection system.

Another study uses data pre-processing operations such as discretization and feature selection to select the necessary features using a feature selection algorithm. This study uses classifiers such as Naive Bayes, NB Tree and Hidden Naive Bayes (HNB) to improve accuracy of the classifier. HNB is used to relax the conditional independence assumption of Naive Bayes method, while NBTree creates a mix of decision tree classifiers and Naive Bayes classifiers to increase accuracy and lower error rates [30]. Another study proposes a model for an Intrusion Detection System using unsupervised machine learning and k-means. The study used NSL-KDD dataset, which has 25,192 entries and 22 types of data. The study shows that better results are produced when the number of data types matches the number of clusters in the dataset [31]. Another study analyzes the NSL-KDD dataset to investigate how well different classification algorithms can identify anomalies in network traffic patterns. The authors also examine how protocols are accessible in widely used network protocol stack relate to the intrusion techniques employed by intruders to produce atypical network traffic. The data

mining tool WEKA's classification methods are used for the analysis. Numerous details regarding the relationship between network assaults and protocols have been revealed by the study [32]. The decision tree method used in the proposed study is based on the C4.5 decision tree approach. When building a decision tree, split value and feature selection are crucial considerations. The proposed Decision Tree Split (DTS) algorithm in this research is created to deal with these two problems. Utilizing information gain, the most important characteristics are chosen, and the split value is chosen to ensure the classifier is unbiased towards the most common values. Analyses are done on the classifier's model construction time and accuracy rates [33]. Table 1 represents the methodology used by different authors with their accuracy.

The authors used several supervised machine learning algorithms, including Logistic Model, Random Forest or Random Tree to forecast the accuracy of an intrusion detection system. The paper likely splits the data into three categories for analysis and evaluates which algorithm performs best in terms of accuracy [34]. This paper proposed a fault prediction recommender model that uses machine learning techniques to predict faults in smart offices. Specifically, it uses

Table 1 Comparison of existing models

Ref.	Methodology	Accuracy (%)
[20]	Random Forest	93.77
	J48	93.10
	Random tree	90.57
	Decision table	92.44
	MLP	91.90
	Naive Bayes	91.23
	Bayes Network	90.73
[22]	J48	99
	SVM	98
	Naive Bayes	93
	Decision Table	99
[24]	XGBoost	98.70
	Decision Table	97.50
	NB	86.20
[26]	Random Forest	99.10
	J48	78.90
	SVM	98.10
	CART	88.90
	NB	70.30
[28]	Decision Tree	79.71
	Random Forest	76.64
	kNN	75.51
	LR	73.58
	SVM	74.09
	DNN	81.60
	Adaboost	76.02
[30]	Naive Bayes	88.20
	Hidden Naive Bayes	93.40
	NBTree	94.60
[31]	k-means clustering	81.61 (for cluster size 22)
[32]	J48	99.10
	SVM	98.70
	Naïve Bayes	75.20
[33]	CART	66.77
	C4.5 Decision Tree	65.65
	Decision Tree Split	79

K-Nearest Neighbor Decision Tree, Gaussian Naive Bayes, or Random Forest and Random Forest approaches to evaluate the model's performance.

The results suggested that RF has maximum accuracy among the classifiers evaluated and also mention that various performance parameters, including F1 score, recall, precision or accuracy were used to assess the performance of model [35–37]. The authors discussed the deployment of ML and deep learning-based IDS to detect network intrusions effectively. However, the paper also highlights some of the shortcomings of the proposed methods and identifies various research challenges in the field. This paper may provide future research directions to improve machine learning and deep learning-based NIDS [38–40]. Some possible areas for improvement could include developing new algorithms that can handle large amounts of data efficiently, improving the accuracy of predictions, and addressing issues related to data privacy and security. Additionally, the paper highlights the need for more research on the interpretability and explainability of machine learning and deep learning models to gain insight into the reasoning behind the predictions and improve trust in the system [41].

The advent of Bitcoin in 2008 by Satoshi Nakamoto [42] has generated significant attention towards blockchain, the foundational technological framework supporting Bitcoin. A blockchain refers to an ever-expanding collection of documents, known as blocks that are interconnected and safeguarded by cryptographic techniques. In blockchain network, the addition of a new block necessitates the approval of all nodes through verification. Blockchain, being a novel and essential technical framework, exhibits suitability for the use of securing cyber-physical systems. The predominant utilization of blockchain technology at present lies within the realm of data security, whereby its primary function is to guarantee the traceability and resistance to tampering of data. For instance, previous research has shown that the combination of blockchain, smart contracts, and Internet of Things technology yields significant benefits [43]. Blockchain technology has the potential to provide durable and genuinely distributed peer-to-peer systems by enabling trustless and auditable interactions among peers. The blockchain technology plays a pivotal role in the registration and authentication of all activities conducted on Internet of Things devices' data [44]. The authors of the study investigate the use of a consortium blockchain in the development of a secure energy trading system referred to as "energy blockchain" [45]. Various research endeavors employ the consensus process as a means to execute distributed and collaborative scheduling decisions or intrusion detection. A proposal is made for the implementation of a completely distributed access control system based on blockchain technology to manage the vast number of devices deployed globally in the Internet of Things ecosystem [46, 47]. This system aims to facilitate the arbitration of roles and permissions for these devices. The potential future operation of the Energy Internet is expected to be increasingly decentralized and self-executing. To facilitate this, it has been proposed that blockchain technology be incorporated into the operational framework of the Energy Internet [48]. This has the potential to foster a consensus among the decentralized institutions of diverse energy organizations. The authors propose a protective framework that utilizes a distributed blockchain technology to augment the self-defensive capabilities of contemporary power systems in countering cyber-attacks [49]. Furthermore, the authors of [50] present a comprehensive analysis of a blockchain framework in the context of IDS. The authors introduced a clustering method that utilizes the principle of gravitation to quantify the extent of clustering irregularity. In order to achieve this objective, a straightforward and efficient approach is devised for determining the clustering threshold, and a unique network intrusion detection technique is formulated using this threshold [51].

The aforementioned methodologies exhibit a high degree of precision in identifying intrusions. However, there is a need for more enhancements to enhance their dependability and efficiency. In summary, blockchain and machine learning technology has been implemented across various domains thus far. This prompts us to examine the application of this technique in the security of IDS systems through the utilization of consensus and incentive mechanisms [52, 53]. Just like the manner in which transaction information is saved in bitcoin, the outcomes of IDS detection with consensus will also be stored on the blockchain. At last, these studies demonstrate the potential of ML techniques and algorithms to enhance the accuracy of IDS and the need to do more work in the intrusion detection field.

3 Materials and methods

One method for assessing system security is the use of IDS, which alerts users when an intrusion is found. Traditional approaches include disadvantages such as false positives and computational expenses, which the suggested ensemble model tries to address with more accuracy and efficiency. In this study, the benchmark KDD'99 Dataset was used which is a publicly available dataset for modern-day internet traffic [34]. The origin of KDD dataset, which is a widely used dataset for research on intrusion detection systems was created as a result of the KDD Cup competition held in 1999, which

aimed to build a predictive model to differentiate between normal connections and intrusions. To create this dataset, a large amount of internet traffic was recorded and collected. The dataset comprises four distinct types of attacks, namely probing attacks that focus on gathering information, Denial-of-Service (DoS) attacks that aim to prevent legitimate requests from reaching a system. The other two attacks are user-to-root (U2R) attacks that involve gaining unauthorized entry to a root or local super-user and remote-to-local (R2L) attacks that involve unauthorized local access from a remote device. KDD dataset has become a standard benchmark dataset for evaluating intrusion detection systems. Many studies have used this dataset to develop and test different ML algorithms and techniques for intrusion detection. The different categories of attacks in the dataset have also been used to study and understand the different types of attacks and their characteristics. Initial steps involve pre-processing and analysis of the KDD dataset to attain main features, characteristics, and visualization of statistical measures which will help us to understand the data at a deeper level. The next process is to develop classifiers based on Gaussian Naive Bayes, XG Boost, and Decision Tree and the final step is to use an ensemble technique to categorize the network data traffic as intrusions and normal connections. The steps involved in various stages of development of this model are described below:

1. Preprocessing: Data preprocessing plays a crucial role in developing an effective IDS model, and it involves several steps such as data cleaning, data transformation, data reduction and data integration. Classification and labeling of data are also essential steps in this process. The main objective of an IDS is to detect and prevent unauthorized access or attacks on a system while minimizing false alarms and suppressing or mitigating DDoS attacks. Therefore, developing an effective IDS model requires careful consideration of various factors, including the selection of appropriate algorithms and features, optimization of parameters, and tuning of thresholds. The KDD dataset has redundant records which cause the biasing of the learning algorithm towards the repetitive values. Also, it has a very less number of records for the R2L class and enormous records for the DDoS class, which will lead to biasing.

   Additionally, the dataset contained a variety of formats, including continuous, discrete, and symbolic data with varying ranges and resolutions. As a result, preprocessing of the data was required before creating a pattern classification model.
2. Exploratory data analysis (EDA): EDA involves using statistical graphics and other data visualization techniques to visualize a dataset to highlight its key features. With a deeper knowledge of the dataset, impartial experiments can be extracted.
3. Building Classifiers:
   a. Decision Tree: Due to its ease of use and simplicity, decision trees are particularly popular as single classifiers.
   b. Gaussian Naive Bayes: Gaussian is a variant of NB that manages continuous data and adheres to Gaussian normal distribution. A set of supervised ML classification methods recognized as Naive Bayes are based on the Bayes theorem. In classification when there are simpler relations, it delivers good results.
   c. XGBoost: The ensemble-based method includes the boosting technique known as eXtreme gradient boosting (XGBoost). The sequential ensemble technique, often known as a sequential decision tree, is created. Because the model is better able to fit the training data, this strategy produces a result that has low bias and good invariance.
4. Classification using ensemble techniques: Using ensemble machine learning to classify network traffic data as legitimate or malicious is a technique that combines numerous learners to work on a given task and increase the precision of classifiers. The maximum voting technique is applied which is frequently employed for categorization issues. With this method, predictions are made for each data point using a variety of models. Each model's predictions are regarded as a "vote." The majority of the models' forecasts serve as the basis for the final projection.

Figure 1 graphically explains the proposed methodology of research work.

4 Results and discussion

In this paper, the authors solved the intrusion detection problem as a binary classification problem, where points belonging to the class "Normal" as one class (Positive Class) and points belonging to the remaining 22 classes as the second class (Negative Class). The reason for converting this problem into a binary classification problem is that organizations are more concerned about Normal and Bad connections getting classified correctly rather than each of the bad categories

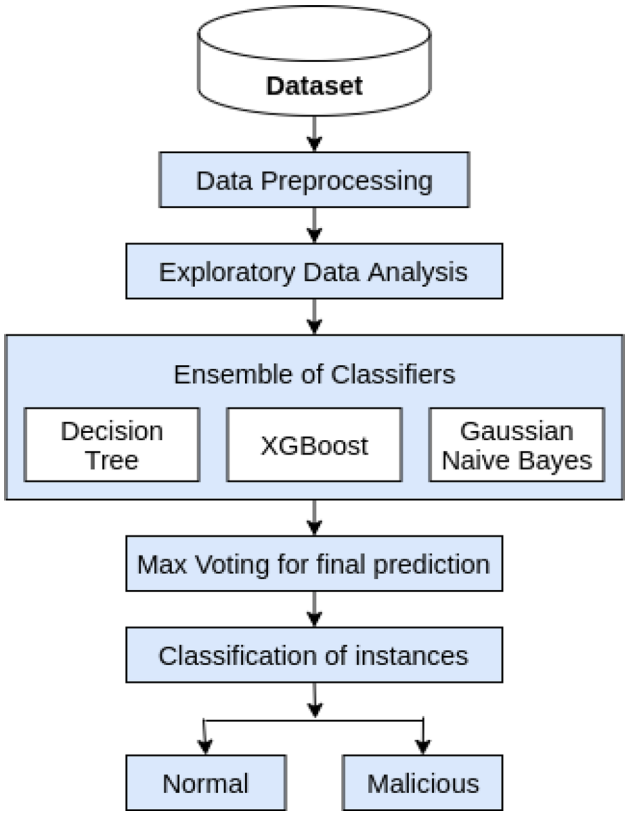


Fig. 1 Proposed methodology for intrusion detection

getting misclassified so that no Bad connections are allowed to gain access to the internal network of the organization by getting misclassified as a normal connection which may otherwise result in a security threat.

4.1 Data preprocessing

Data preprocessing is the process of getting the raw data ready (cleaning and organizing) so that it can be used to create and train machine learning models. It checks for and eliminates duplicate rows and null values as shown in Fig. 2. One cannot work with raw data makes it a crucial step as well. Before using machine learning techniques, the data's quality should be examined. Preprocessing of data increases the reliability as well as the accuracy of an IDS model. Elimination of inconsistent or missing data values occurred by computer or human errors, preprocessing can make the dataset more dependable, and consequently, the model more accurate. Moreover, preprocessing techniques such as feature and instance selection can reduce the dimensionality of the dataset, making it easier to process and analyze. Data transformation techniques such as converting class to nominal and symbolic features to numeric can also help to improve the performance of the model by standardizing and normalizing the data. Overall, preprocessing is an essential step in the development of an effective IDS model that can accurately detect and prevent unauthorized access or attacks on a system.

4.2 Exploratory data analysis

By using summary statistics and graphical representations, Exploratory Data Analysis (EDA) offers a thorough comprehension of the dataset to uncover patterns and test hypotheses.

The dataset used for intrusion detection system contains 23 output labels or categories of network traffic, including 'normal', which constitutes about 60.33% of the data points. The remaining categories represent different types of attacks on the network. Among them, 'neptune' has the highest number of data points with 35.594%, followed by 'back' with 0.665%. This indicates that dataset is highly imbalanced and it is important to build a model that can accurately classify data points from the low distribution classes and identify intrusion categories accurately. The information presented in Fig. 3 is visually represented in Fig. 4.

Fig. 2 Data cleaning

```
# Checking for null values
for index, value in df.isnull().sum().iteritems():
    if value > 0:
        print(f"There are {value} missing values in column - {index}")
```

```
[12] # There are no null values
     # Checking for duplicate rows
     print(f"Duplicate rows - {df.duplicated().sum()}")
```

```
Duplicate rows - 348435
```

```
[13] # Removing duplicate rows
     df.drop_duplicates(keep='first', inplace=True)
     print(f"Duplicate rows - {df.duplicated().sum()}")
```

```
Duplicate rows - 0
```

```
[14] # Saving the cleaned dataset
     df.to_pickle('clean_dataset.pkl')
```

Fig. 3 Target class distribution

```
[17] # Distribution of classes in target label
     dist = df['intrusion_type'].value_counts()
     for name, count in dist.iteritems():
         print(f"Number of data points in class: {name.center(17)} is", end = ' ')
         print(f"{count} ({np.round(count / sum(dist) * 100, 3)}%)")
```

```
Number of data points in class:      normal.      is 87832 (60.33%)
Number of data points in class:      neptune.     is 51820 (35.594%)
Number of data points in class:       back.       is 968 (0.665%)
Number of data points in class:     teardrop.     is 918 (0.631%)
Number of data points in class:       satan.      is 906 (0.622%)
Number of data points in class:    warezclient.   is 893 (0.613%)
Number of data points in class:      ipsweep.     is 651 (0.447%)
Number of data points in class:       smurf.      is 641 (0.44%)
Number of data points in class:     portsweep.    is 416 (0.286%)
Number of data points in class:        pod.       is 206 (0.141%)
Number of data points in class:       nmap.       is 158 (0.109%)
Number of data points in class:   guess_passwd.   is 53 (0.036%)
Number of data points in class:  buffer_overflow. is 30 (0.021%)
Number of data points in class:    warezmaster.   is 20 (0.014%)
Number of data points in class:       land.       is 19 (0.013%)
Number of data points in class:       imap.       is 12 (0.008%)
Number of data points in class:      rootkit.     is 10 (0.007%)
Number of data points in class:    loadmodule.    is 9 (0.006%)
Number of data points in class:     ftp_write.    is 8 (0.005%)
Number of data points in class:     multihop.     is 7 (0.005%)
Number of data points in class:        phf.       is 4 (0.003%)
Number of data points in class:       perl.       is 3 (0.002%)
Number of data points in class:        spy.       is 2 (0.001%)
```

Heatmaps are useful for visualizing patterns and relationships within a dataset. They are commonly used in exploratory data analysis to identify correlations between variables or to identify outliers or unusual values. Heatmaps can also be used for clustering and classification analysis, as well as for visualizing time-series data. Overall, heatmaps provide a powerful tool for analyzing and visualizing complex data sets. The feature to feature relationship is analyzed and presented through heat map in Fig. 5.

Pair plots shown in Fig. 6 are a useful visualization tool for exploring the pairwise relationships between multiple variables in a dataset. They allow us to quickly identify patterns and correlations between variables, as well as potential

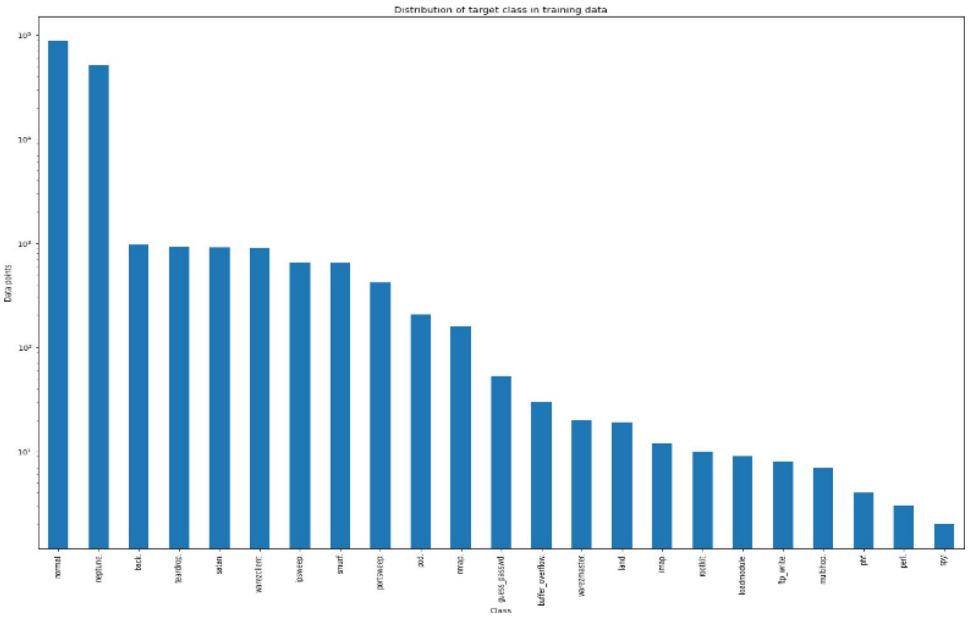


Fig. 4 Target class distribution during training

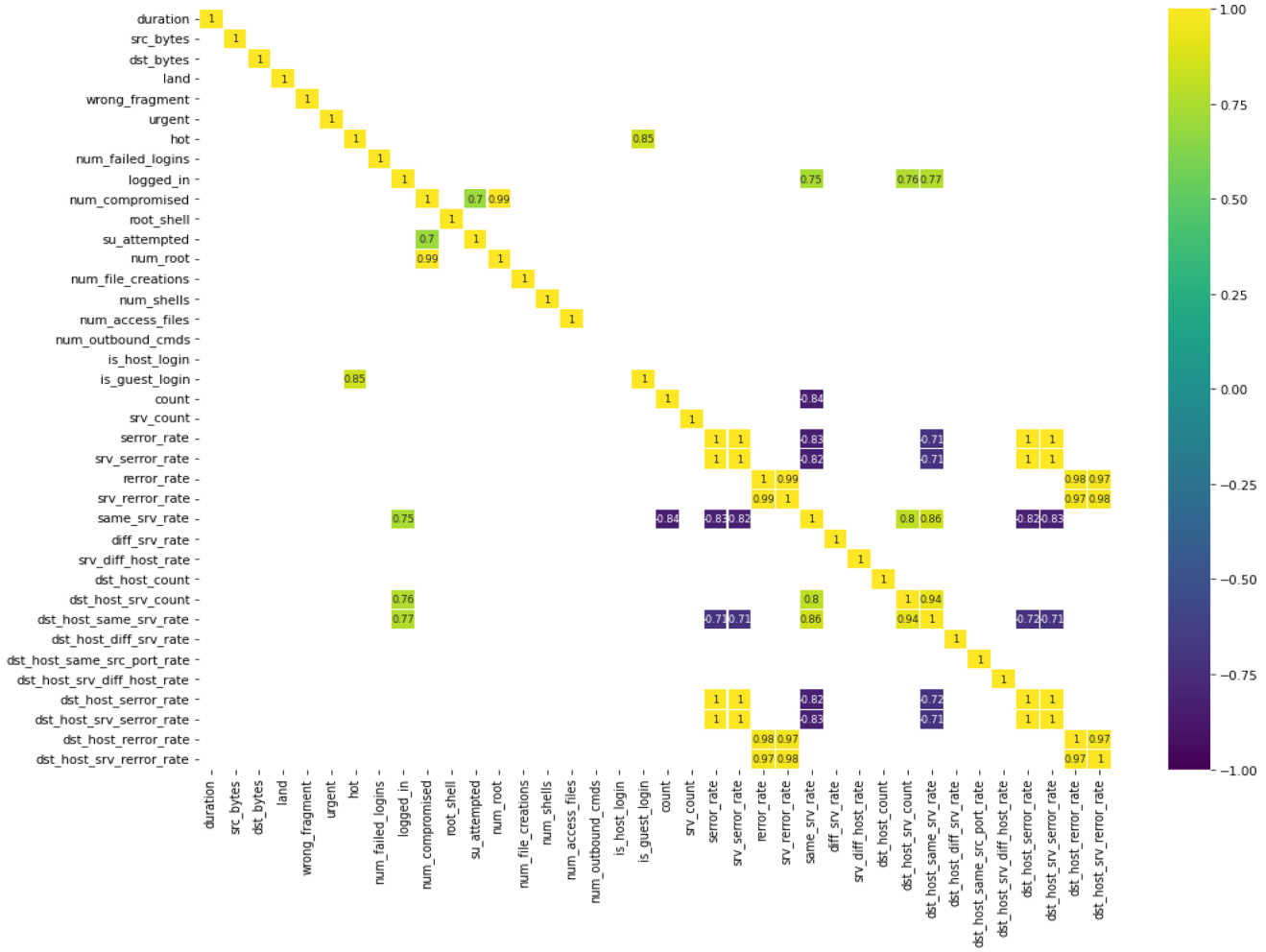


Fig. 5 HeatMap representing correlation values for the input features

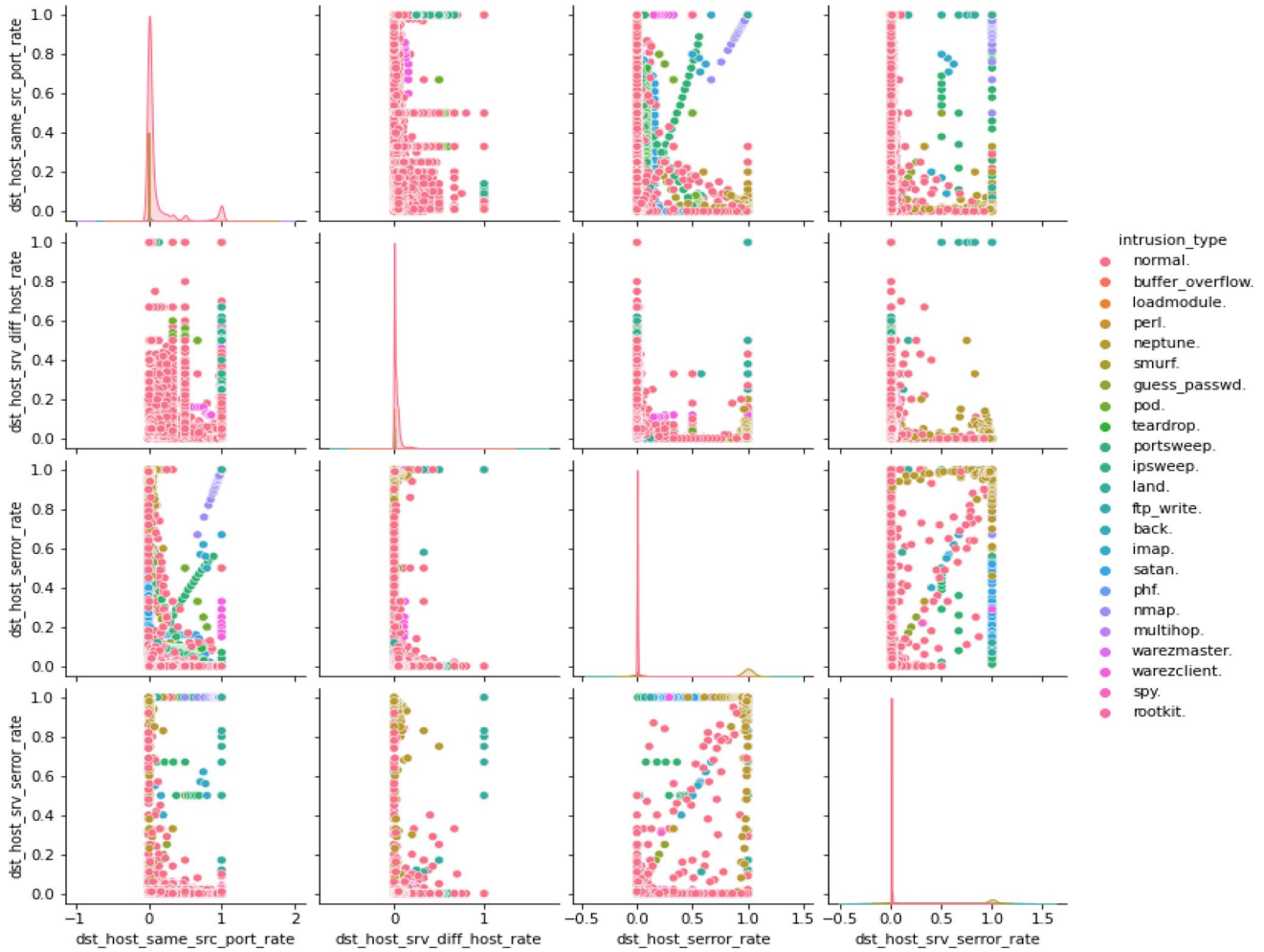


Fig. 6 Pairplot of variables: 'dst_host_serror_rate', 'dst_host_same_src_port_rate', 'dst_host_srv_serror_rate', 'dst_host_srv_diff_host_rate'

outliers or anomalies. By plotting each pair of variables against each other in a scatter plot format, pair plots can help us to identify clusters or groups within the data as well as detect any nonlinear relationships or interactions between variables. Pair plots can also be used to identify potential confounding variables that may need to be controlled for in subsequent analyses. Seaborn is a popular Python library that provides functions for creating pair plots, and it offers a range of customization options to help tailor the plot to specific needs.

Principal Component Analysis (PCA) technique reduce the number of features in a dataset while maintaining variance to a great extent. It transforms the original data into a new coordinate system by determining the directions (principal components) with highest variance and then, on these directions, data is projected. The resultant principal components are uncorrelated and orthogonal to each other. The training data set's PCA graphic in Fig. 7 shows that the Normal (pink), Neptune (mustard), and phf (blue) data are highly centered and have little overlap with other types of data, making them simple to categorize. Machine learning algorithms can be employed for training after the training data set has been preprocessed. PCA can be used for data visualization, noise reduction and feature extraction. In the context of IDS, PCA can help identify the most important features for detecting attacks and reduce the computational complexity of the algorithms used for classification as shown in Fig. 8.

It is observed that TCP has the highest number of good and bad connections among the given data set. There is a very less number of cases where a root shell has been obtained. This is because the root shell is generally used by system administrators only. For attackers, they need to get access to the box as a user before escalating their privileges to root. It was further observed that whenever there was a case of root shell access, a buffer overflow attack was encountered. The reason being when escalating privileges from user to root, generally only 2 types of attack are possible, misconfiguration in permissions or buffer overflows. Buffer overflows are more common as most Kernel exploits are buffer overflows. In

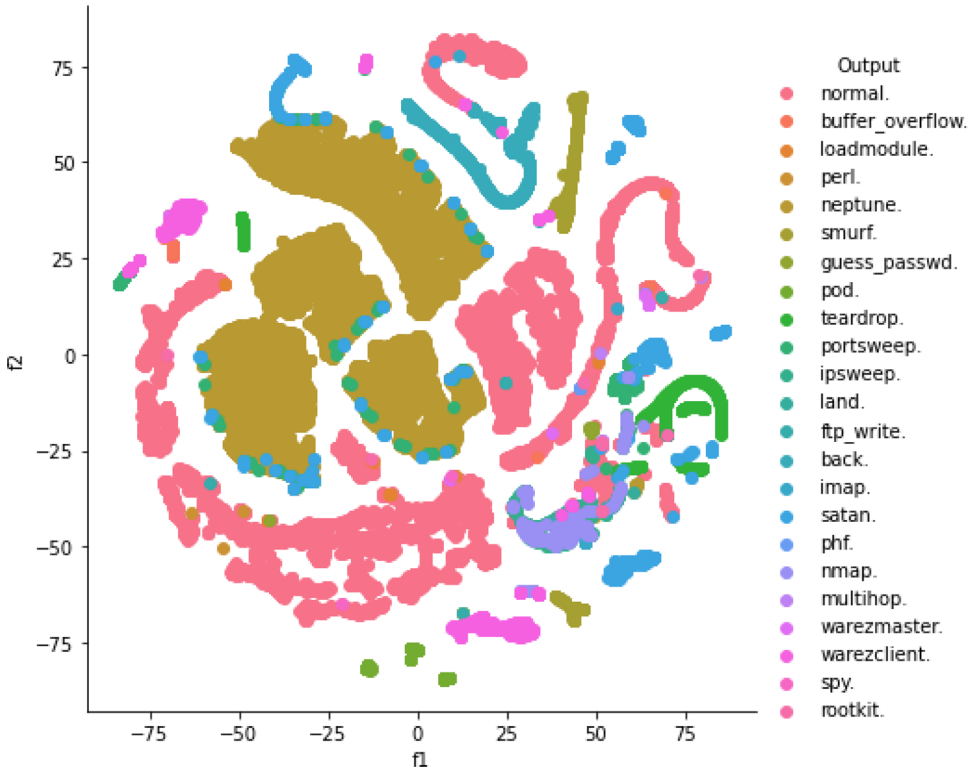


Fig. 7 t-SNE visualization of data

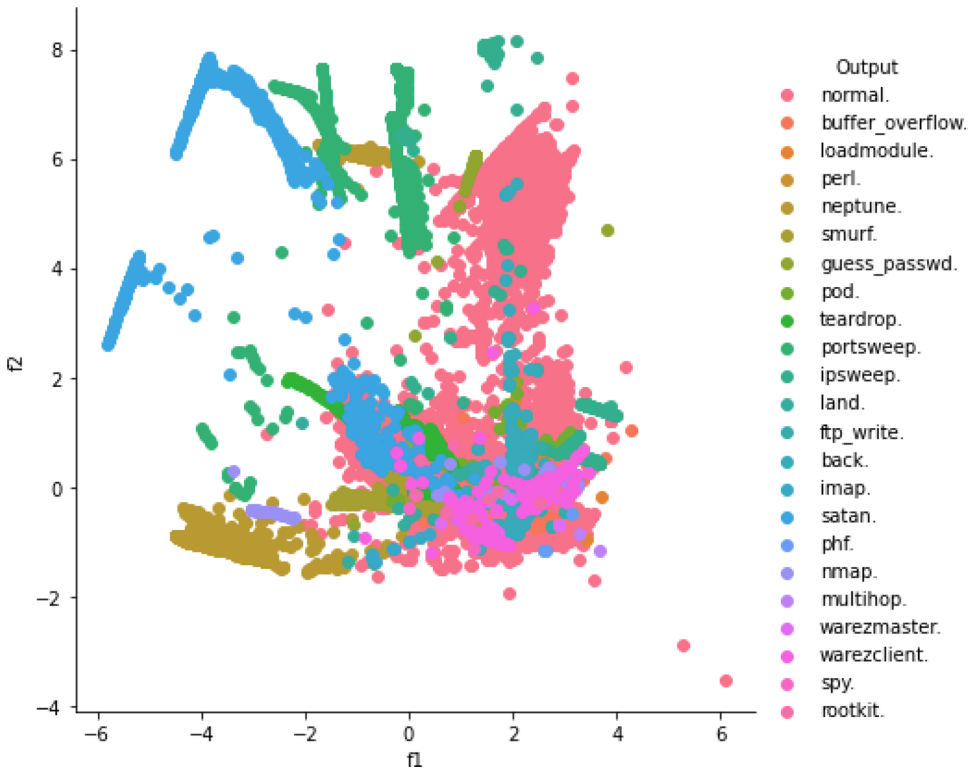


Fig. 8 Principal component analysis of training data set

Neptune attacks, the attacker sends a flood of SYN packets and the target sends back SYN-ACK packets in reply. From this, the attacker comes to know that the target is alive and sends a packet with REJ and S0 flags. Furthermore, there are a lot of packets with SF flags in normal secure connections.

4.3 Model training and testing

a. Gaussian Naive Bayes: Gaussian Naive Bayes assumes that the input features are continuous and follow a Gaussian (normal) distribution. It also assumes that the features are independent of each other, meaning that the presence or absence of one feature does not affect the probability of the other features. It is a probabilistic algorithm that uses the Gaussian distribution to model the probability distribution of each class. To make a prediction using Gaussian Naive Bayes, the algorithm first calculates the probability of each class given the input features using Bayes' theorem. Then, it selects the class with the highest probability as the predicted class. One advantage of Gaussian Naive Bayes is its simplicity and efficiency, as it can be trained quickly with relatively little data. Additionally, it may not be suitable for categorical or ordinal features, which have discrete values.
b. Decision Tree: In a decision tree, the process of dividing the root node into subsets is based on finding the attribute that provides the most information gain or the greatest reduction in impurity. This process is repeated on each resulting subset, creating a tree-like structure until a stopping criterion is met, like lowest number of instances in leaf node, maximum tree depth or lowest information gain. Each internal node in the decision tree splits the data into two or more subsets based on the attribute's values. The branches represent further nodes or leaf nodes. The leaf nodes represent classification or regression output.
c. XGBoost: XGBoost is an ensemble method that trains a series of weak decision trees and combines their predictions to produce a more accurate overall prediction. It uses gradient boosting, which is a machine learning technique that iteratively trains weak models by focusing on the samples that were misclassified by previous models and adjusting their weights to minimize the error. XGBoost is known for its high performance and accuracy and has been widely used in various machine learning competitions and real-world applications.

Confusion matrix evaluates the performance of a classification model, particularly in cases where there are multiple classes being predicted. It permits watching how many true negatives, true positives, false negatives, and false positives the model has made, which in turn calculates a variety of evaluation metrics such as precision, accuracy, specificity, recall, and sensitivity. Additionally, the information in the confusion matrix can be used to generate a receiver operating characteristic (ROC) curve that shows a trade-off between a false positive rate and a true positive rate for distinct classification thresholds. Some of the fundamental terms are listed below:

- True Negative (TN): The model predicted a negative class when the actual class is also negative.
- False Negative (FN): The model predicted a negative class but the actual class is positive.
- True Positive (TP): The model predicted a positive class when the actual class is also positive.
- False Positive (FP): The model predicted a positive class but the actual class is negative.

The confusion matrix of Decision Tree, Gaussian Naive Bayes and XGBoost are represented in Fig. 9.

The performance of a classifier for each potential threshold is shown on a graph called the ROC curve which is a graphical representation of binary classifier system's performance. The ROC curve can be used to evaluate the performance of different classification models by comparing their AUC (Area Under the Curve) values. The AUC represents the degree of separability between the classes, which indicates how well the model is able to distinguish between positive and negative classes. A model with a higher AUC value is considered to have better performance.

In Fig. 10, the ROC curves of Gaussian Naive Bayes, Decision Tree, and XGBoost are shown, allowing for a comparison of their performance in terms of true positive rate and false positive rate. The curves demonstrate how well each model can differentiate between the two classes, with the highest AUC value indicating the best overall performance.

4.4 Proposed ensemble model

An ensemble model proposed in this paper selects well-known machine learning algorithms as alternative classifiers as shown in Fig. 11. These include Gaussian Naive Bayes, Decision Trees and XGBoost. Comparative experiments are used to choose three voting classifiers. To get the best results, the max voting technique is proposed. The dataset for intrusion detection is used to test an ensemble-based machine learning model that combines the Gaussian Naive Bayes classifier, Decision Tree classifier and XGBoost classifier. On the KDDCup99 dataset, data preprocessing is first carried out by removing duplicate rows and null values. On this normalized dataset, exploratory data analysis is then performed, and a proposed ensemble model is trained. Finally, the performance of the suggested model is assessed in comparison to

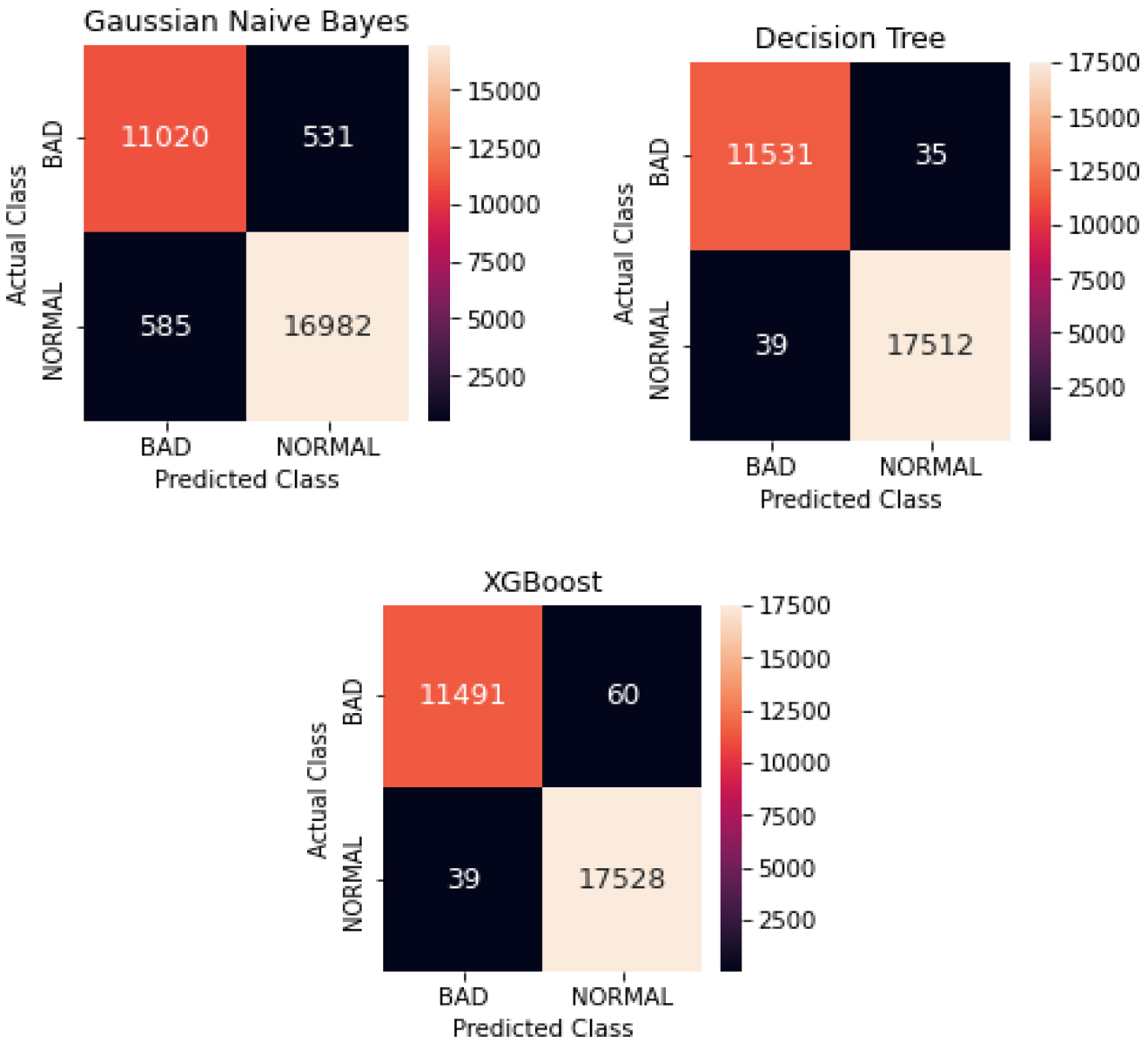


Fig. 9 Confusion matrix of Decision Tree, Gaussian Naive Bayes and XGBoost

the various ML techniques. According to comparison analysis, the proposed model outperforms the individual machine learning methods.

Multiple techniques can be combined to produce better results when voting compared to the single model, it can produce results that are more precise. This technique has been applied in various industries as a categorization approach to boost performance. Instead of using a single model in this research project, the voting process obtains more accurate results. Any ML approach, such as Decision Tree, Gaussian Naive Bayes and XGBoost can be used as the sole model. In the context of ensemble learning, the base model is typically the individual model that performs the best on the training data or has the highest accuracy. The idea behind using a max-voting ensemble method is to combine the predictions of multiple models to improve the overall accuracy and reduce the risk of overfitting. Max voting, or hard voting, is a simple method where each model receives one vote and the class with maximum votes is chosen as the final prediction. Soft voting, on the other hand, takes into account the probabilities of each class predicted by each model and calculates the average probability for each class. The class with the highest average probability is then chosen as the final prediction. Soft voting can be more effective than hard voting when the individual models have high variance or are prone to overfitting. The confusion matrix and ROC curve of proposed technique is shown in Fig. 12.

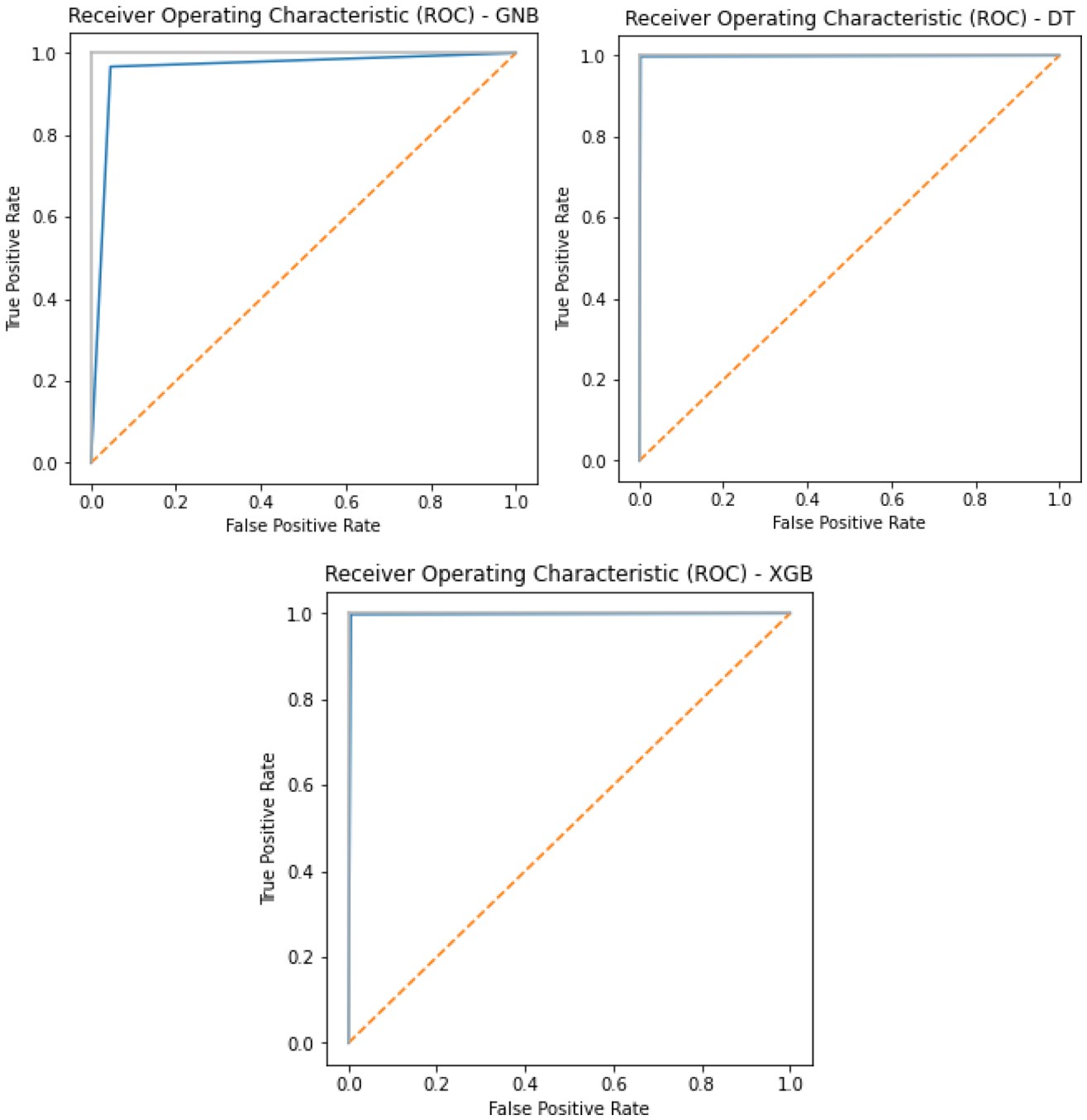


Fig. 10 ROC curves for Decision Tree, Gaussian Naive Bayes and XGBoost

When compared to other classification models, XGBoost consistently offer greater prediction accuracy. After organizing the proposed ensemble method, the overall prediction accuracy of this study came out to be 99.86%, which is considered a pretty required accuracy. The proposed approach is evaluated using six evaluation metrics: accuracy, precision, recall, f1-score, specificity and sensitivity. Each model performed differently on different evaluation metrics as shown in Fig. 13.

The graph of comparison of accuracy, precision, recall, f1-score and error rate with different classifiers such as Gaussian naive Bayes, decision tree and XGBoost and proposed ensemble technique are represented in Table 2. After applying Gaussian naive Bayes, an accuracy of 96.62%% and an error rate of 3.38% were obtained. In the decision tree classifier, the performance has improved in terms of accuracy. In the case of XGBoost, the accuracy raised from 96.62 to 99.67% and the error rate dropped from 3.38 to 0.33%. The proposed ensemble technique has performed best and has a maximum accuracy 99.88% and an error rate 0.12%. The improved performance of this study compared to previous findings

```
# Vote from GNB
df = pd.read_pickle('gnb_dataset.pkl')
preds = []
test_set = df.iloc[int(np.ceil(.80 * df.shape[0])):, :-1]
for test in test_set.values:
    preds.append(gaussian_classifier.predict(test))
gnb_preds = np.array(preds)

# Vote from DT
df = pd.read_pickle('dt_dataset.pkl')
test_set = df.iloc[int(np.ceil(.80 * df.shape[0])):, :].values.tolist()
predicted = []
for row in test_set:
    predicted.append(sorted(classify(row, my_tree).items(), key = lambda x: x[1]
dt_preds = np.array(predicted)

# Vote from XGBoost
df = pd.read_pickle('xgb_dataset.pkl')
test_set = df.iloc[int(np.ceil(.80 * df.shape[0])):, :-1]
xgb_preds = model.predict(test_set)

# Max-voting ensemble technique
y_test_pred_maxvote = []
for tup in list(zip(gnb_preds, dt_preds, xgb_preds)):
    y_test_pred_maxvote.append(np.bincount(tup).argmax())
max_vote_preds = np.array(y_test_pred_maxvote)

# Evaluation
y_test = df.iloc[int(np.ceil(.80 * df.shape[0])):, -1]
```

Fig. 11 Proposed ensemble technique

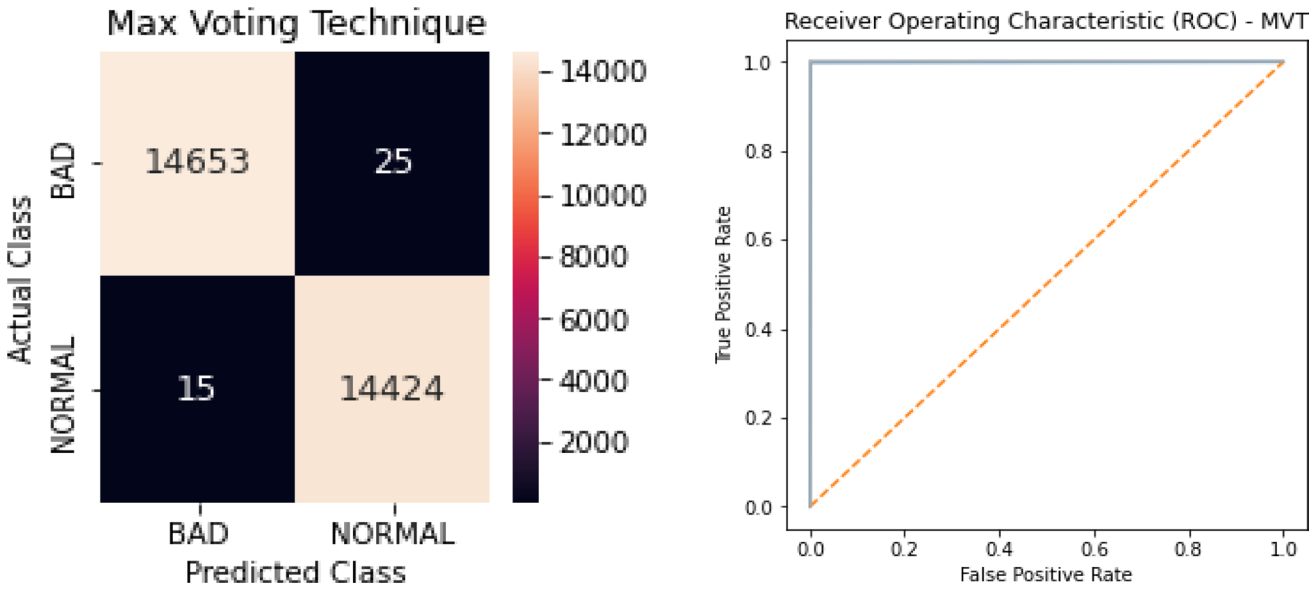


Fig. 12 Proposed ensemble technique results

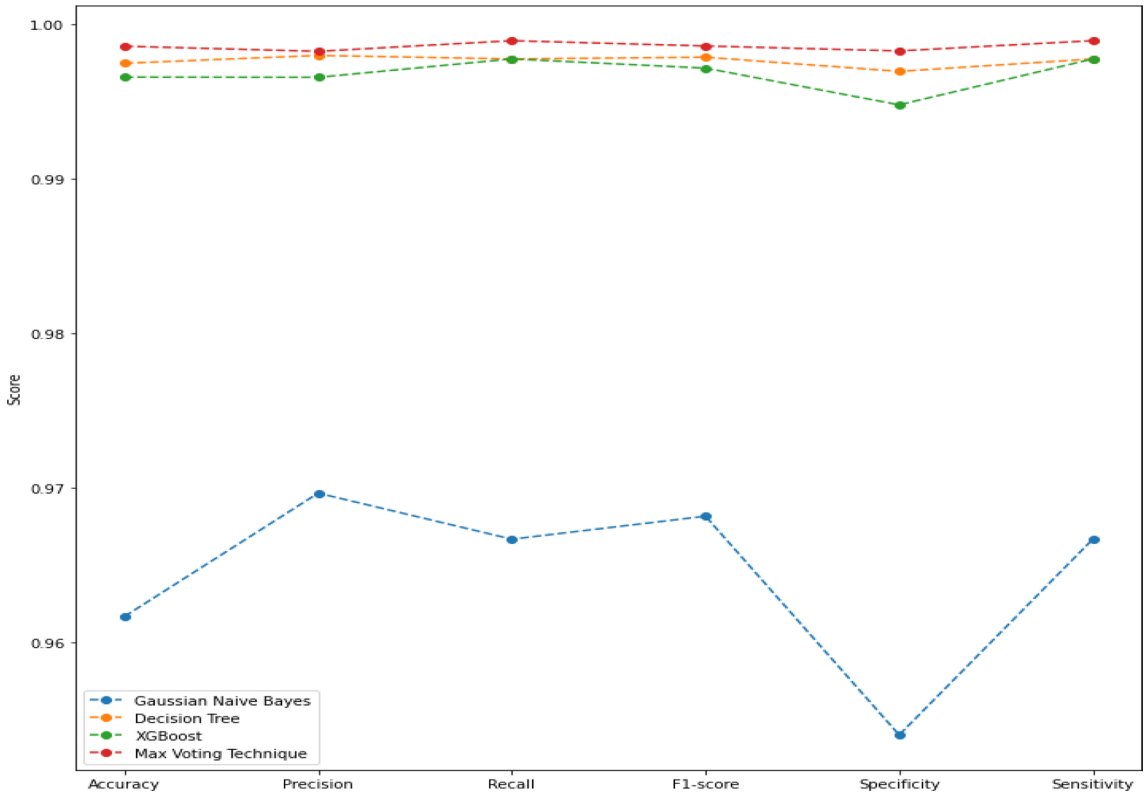


Fig. 13 Comparison of Gaussian Naive Bayes, Decision Tree and XGBoost and proposed ensemble technique

Table 2 Comparison of Accuracy and error rate of proposed model with different classifiers

Technique	Accuracy (%)	Error rate (%)	Precision (%)	Recall (%)	F1-Score (%)
Gaussian Naive Bayes	96.62	3.38	96.8	96.5	96.65
XGBoost	99.65	0.35	99.7	99.6	99.65
Decision Tree	99.67	0.33	99.7	99.7	99.7
Proposed Technique	99.88	0.12	99.9	99.85	99.87

can be due to the successful attainment of higher levels of accuracy and a significantly reduced error rate within the domain of intrusion detection, achieved through the utilization of an ensemble-based machine learning model. The empirical evaluation presented in Table 2 provides strong evidence that the suggested methodology outperforms all already available methodologies. Figure 14 highlights how changes in a single measurement correspond with changes in another, giving a far reaching perspective on every method's performance. The accuracy is addressed by bars, making it simple to look at across methods, while the error rate, plotted as a line with markers, offers knowledge into how every method's accuracy conversely connects with its error rate. To improve the IDS's resilience and generality, ensemble learning methodologies such as maximum voting were employed. By evaluating models such as XGBoost, GNB and DT, the researchers discovered that integrating predictions with maximum voting increases accuracy and reduces error rates. Ensemble learning enhances expected performance by combining the capabilities of individual models while addressing each model's deficiencies. The ensemble model outperformed the single models in the study, with an accuracy of 99.88%. This method was chosen above others because it provides a more accurate and balanced categorization, especially in complex intrusion detection scenarios.

Figure 15 shows Precision-Recall (PR) curve for every method: Gaussian Naive Bayes, XGBoost, Decision Tree and the proposed method. It gives a graphical representation of every method's performance as far as how precisely the positive class is anticipated across various threshold settings.

This serves as evidence of the strength and effectiveness of the methodology that was implemented. By employing rigorous research methodology and utilizing sophisticated approaches, this study not only boosts the overall effectiveness of

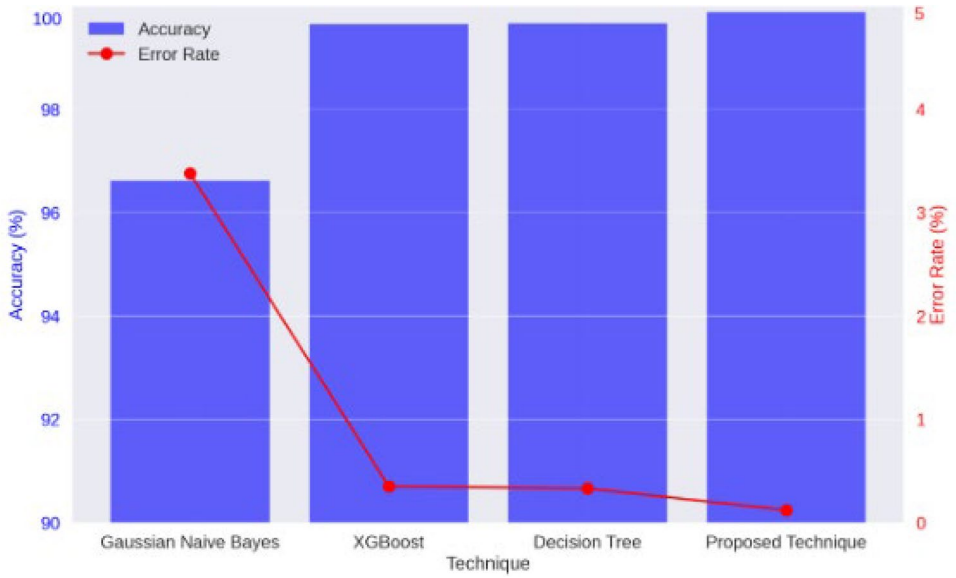


Fig. 14 Accuracy vs error rate for each technique

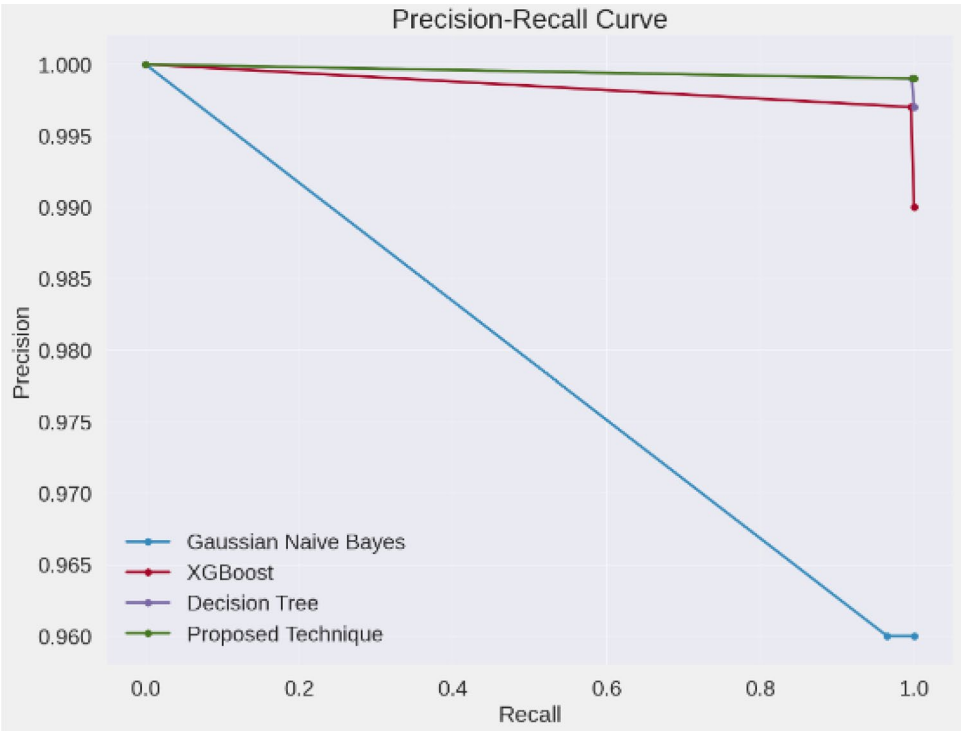


Fig. 15 PR curve comparison of Gaussian Naive Bayes, Decision Tree and XGBoost and proposed ensemble technique

intrusion detection but also improves its practical applicability and reliability. This study provides a more dependable and accurate solution, thereby contributing to the advancement of the field of intrusion detection. In future, the efficiency of the clustering model can be enhanced using a Blockchain network. Due to the diverse nature of network attacks, the utilization of an anomaly detection method is applicable in several scenarios, such as the identification of anomalies within data packets and network traffic.

5 Conclusions and future scope

With the digitization of data and rapid development in the field of information technology, computer networks are used widely in industries and businesses. This advancement in technology has also brought the risk of attacks and threats to computer networks' availability, integrity, and confidentiality. Computer network and network security are the areas that require great attention due to the increase in the number of attacks every day. Blockchain technology enables decentralized transactions and data management without a trusted third party. While blockchains have been researched in healthcare and supply chain management, their potential use in intrusion detection has not been extensively studied. The authors suggested an adaptable ensemble learning model in this study. The proposed model's main concept is to employ an ensemble technique to combine the benefits of many ML techniques. The proposed ensemble technique has performed best which has an accuracy of 99.88% and an error rate is 0.12%.

In future, optimizing feature selection and preprocessing can definitely lead to better results in intrusion detection using machine learning techniques. In addition, exploring other machine learning algorithms or ensemble techniques and experimenting with different hyperparameters can also improve the accuracy and robustness of the model. It is important to constantly evaluate and improve the performance of the model as new data and threats emerge. Further, other combinations of ML and DL models can be used to improve prediction. The integration of blockchain and encryption technologies is a promising avenue for future study in achieving secure identity identification of blockchain data. The enhancement of intrusion detection's real-time performance and security can be seen as a potential benefit of improving the controllability of data anonymity and privacy protection within a Blockchain network. In the area of network security research and application, ensemble machine learning offers a good generalization effect that merits ongoing promotion and optimization.

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Data availability The source of datasets are 1. https://www.kaggle.com/datasets/galaxyh/kdd-cup-1999-data. 2. https://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html.

Declarations

Competing interests The authors declare no competing interests.

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